RAMCO VIDYA MANDIR SR. SEC. SCHOOL, THAMARAIKULAM	
CLASS :	**TIME : 3 Hrs**
EXAM NO :	**MARKS : 80**

Class 11 Biology

Unit: 4

Difficulty: Hard

Total Marks: 100

Duration: 3 hours

SECTION A: MCQs (1 mark each) (14 × 1 = 14)

CBSE Class 11 Biology MCQs

1) Which of the following statements about Phylum Ctenophora is INCORRECT?
a) They possess eight external rows of ciliated comb plates
b) They exhibit both extracellular and intracellular digestion
c) They are exclusively freshwater organisms
d) They display bioluminescence

2) In Ctenophores, the comb plates primarily function as:
a) Defensive structures
b) Organs for bioluminescence
c) Locomotory structures
d) Feeding appendages

3) Assertion: Ctenophora are classified as diploblastic animals.
Reason: They have tissue level of organization.
a) Both assertion and reason are true, and reason is the correct explanation of the assertion
b) Both assertion and reason are true, but reason is not the correct explanation of the assertion
c) Assertion is true but reason is false
d) Both assertion and reason are false

4) Based on the given text, which of the following is a unique characteristic of ctenophores that distinguishes them from cnidarians?
a) Presence of radial symmetry
b) Presence of comb plates for locomotion
c) Diploblastic organization
d) Tissue level organization

5) Which of the following phyla exhibit radial symmetry like the Ctenophora?
a) Porifera and Annelida

b) Coelenterata and Echinodermata
c) Platyhelminthes and Arthropoda
d) Mollusca and Chordata
6) Case study: A marine organism was collected from the ocean and observed to emit light in the dark. It had a translucent, jelly-like body with eight rows of cilia. Which of the following conclusions is correct?
a) It belongs to Phylum Porifera as it has a jelly-like body
b) It belongs to Phylum Coelenterata as it has cnidoblasts
c) It belongs to Phylum Ctenophora as it shows bioluminescence and has eight rows of cilia
d) It belongs to Phylum Echinodermata as it has radial symmetry
7) Assertion: Ctenophores are more advanced than Poriferans.
Reason: Ctenophores show tissue level organization while Poriferans show cellular level organization.
a) Both assertion and reason are true, and reason is the correct explanation of the assertion
b) Both assertion and reason are true, but reason is not the correct explanation of the assertion
c) Assertion is true but reason is false
d) Both assertion and reason are false
8) Which of the following is NOT a characteristic feature of Phylum Ctenophora?
a) Presence of cnidoblasts for capturing prey
b) Presence of diploblastic organization
c) Presence of radial symmetry
d) Absence of separate sexes
9) The locomotory structures in ctenophores are:
a) Flagella arranged in eight rows
b) Ciliated comb plates arranged in eight rows
c) Tube feet arranged radially
d) Parapodia arranged segmentally
10) Case study: A marine biologist found two different organisms - Organism A and Organism B. Organism A has a water canal system with choanocytes, while Organism B has eight rows of ciliated comb plates and exhibits bioluminescence. The correct classification would be:
a) Organism A belongs to Porifera and Organism B belongs to Ctenophora
b) Both organisms belong to Porifera but different classes
c) Both organisms belong to Ctenophora but different classes
d) Organism A belongs to Ctenophora and Organism B belongs to Porifera
11) Assertion: Ctenophores have a more complex body organization than coelenterates.
Reason: Ctenophores have mesoderm while coelenterates do not.
a) Both assertion and reason are true, and reason is the correct explanation of the assertion
b) Both assertion and reason are true, but reason is not the correct explanation of the assertion
c) Assertion is true but reason is false
d) Both assertion and reason are false
12) Which of the following features is common to both Ctenophora and Coelenterata?
a) Presence of nematocysts
b) Diploblastic organization

c) Presence of water vascular system
d) Presence of specialized excretory organs
13) The property of a living organism to emit light is called:
a) Bioluminescence
b) Phosphorescence
c) Fluorescence
d) Incandescence
14) In the classification of Kingdom Animalia, Ctenophores are placed:
a) Before Porifera due to their simpler organization
b) After Porifera but before Coelenterata
c) After Coelenterata but before Platyhelminthes
d) After Platyhelminthes but before Annelida

SECTION B: Very Short Answer (2 marks each) (10 × 2 = 20)

15) Analyze how the arrangement of vascular bundles in dicot stems creates a "ring" formation and explain why this structural organization is significant for secondary growth.
16) Compare the anatomical adaptations of bulliform cells in monocot leaves with those of the spongy parenchyma in dicot leaves in terms of their functional responses to water stress.
17) Explain why the casparian strips in the endodermal cells of roots are crucial for the selective absorption of water and minerals from the soil.
18) Differentiate between the structure and function of pericycle in roots versus stems, focusing on its developmental potential.
19) Evaluate how the arrangement of palisade and spongy parenchyma in dorsiventral leaves optimizes photosynthetic efficiency compared to isobilateral leaves.
20) Compare the vascular bundle structure in monocot stems with that in dicot stems, and explain how these differences affect their mechanical strength.
21) Explain how the structural differences between bean-shaped and dumb-bell shaped guard cells influence stomatal function in dicots versus grasses.
22) Analyze why the position and abundance of stomata differ between the adaxial and abaxial epidermis of leaves, relating this to photosynthetic efficiency and water conservation.
23) Describe how the morphological features of a typical flower reflect its evolutionary adaptations for successful sexual reproduction in angiosperms.
24) Compare the structural organization of endospermic and non-endospermic seeds, explaining the physiological significance of these differences during germination.

SECTION C: Short Answer Type I (3 marks each) (7 × 3 = 21)

25) Explain how the structural adaptations of bulliform cells in grass leaves help the plant respond to water stress conditions. Illustrate your answer with specific changes that occur at the cellular level during these responses.
26) Compare and contrast the structural arrangement of vascular bundles in dicot and monocot stems. How do these differences influence the ability of these plants to undergo secondary growth?

27) Analyze the functional significance of the cuticle layer in plant epidermis. How does its presence or absence in different plant parts relate to the specific environmental challenges faced by those structures?

28) Describe how the anatomical features of the endodermis in roots contribute to selective water and mineral absorption. Why is this tissue considered a critical boundary between the cortex and stele?

29) Explain how the arrangement and structure of mesophyll tissue in dorsiventral leaves optimize photosynthetic efficiency. What specific adaptations do these cells show in relation to their position within the leaf?

30) Analyze the relationship between the distribution of mechanical tissues (collenchyma and sclerenchyma) in dicot stems and their functional role in providing structural support. How does their positioning relate to the mechanical stresses experienced by the plant?

31) Examine how the anatomical differences between monocot and dicot roots reflect their different growth patterns and functional requirements. Focus specifically on the organization of vascular tissues and presence of pith.

SECTION D: Case-based/Competency (4 marks each) (2 × 4 = 8)

Case-Based Question 1: Bioluminescent Ctenophores (4 marks)

Marine biologist Dr. Maya Patel is studying bioluminescence in ctenophores (comb jellies). She collects several specimens from different ocean depths and observes their bioluminescent capabilities under laboratory conditions. Her team notices that specimens from deeper waters produce more intense blue-green light compared to those from shallower waters. They also observe that the bioluminescence seems to occur in specific patterns along the eight external rows of ciliated comb plates.

Dr. Patel's team conducts an experiment where they expose the ctenophores to different stimuli and record the intensity of bioluminescence on a scale of 1-10:

Stimulus	Shallow water specimens (0-100m)	Deep water specimens (500-1000m)
Touch	3	8
Light	1	7
Chemical	4	9
Sound	0	0

Based on the above information, answer the following questions:

a) Explain why the deep-water ctenophores might exhibit more intense bioluminescence compared to shallow-water specimens. (1 mark)

b) The team observes that ctenophores display bioluminescence when predators approach. What evolutionary advantage might this provide to these diploblastic organisms? (1 mark)

c) Ctenophores exhibit both extracellular and intracellular digestion. How might their bioluminescent capabilities relate to their tissue level of organization? (1 mark)

d) If Dr. Patel were to examine the reproductive system of these ctenophores, what would she expect to find regarding their sexual characteristics? How might this affect the distribution of bioluminescent traits in their population? (1 mark)

Case-Based Question 2: Leaf Anatomy and Environmental Adaptation (4 marks)

Botanist Dr. Raj Sharma is studying how plants adapt to different environmental conditions through modifications in their leaf anatomy. He collects leaf samples from various plant species growing in three different habitats: desert, tropical rainforest, and aquatic environment. He prepares transverse sections of these leaves and examines them under a microscope.
The following observations were recorded:
Sample A (Desert plant):
- Thick cuticle on both surfaces
- Sunken stomata primarily on lower epidermis
- Compact mesophyll with minimal intercellular spaces
- Bulliform cells present
- Small, densely packed vascular bundles
Sample B (Tropical rainforest plant):
- Thin cuticle
- Numerous stomata on lower epidermis
- Well-differentiated palisade and spongy parenchyma
- Large intercellular spaces in spongy parenchyma
- Well-developed vascular bundles
Sample C (Aquatic plant):
- Very thin or absent cuticle
- Stomata on upper epidermis only
- Large air cavities throughout mesophyll
- Reduced vascular tissue
- Chloroplasts concentrated in upper epidermis
Based on these observations, answer the following questions:
a) Which anatomical features in Sample A represent adaptations to prevent water loss, and how do they function? (1 mark)
b) Compare and contrast the arrangement of stomata in Samples A and C. How does this arrangement relate to their respective environments? (1 mark)
c) If you were to examine the transverse section of Sample B further, what would you expect to observe about its palisade parenchyma cells compared to those in Sample A? Explain your reasoning based on their different habitats. (1 mark)
d) Sample C shows significantly reduced vascular tissue compared to the other samples. Explain how this anatomical feature relates to the plant's habitat and method of obtaining water and nutrients. (1 mark)

SECTION E: Long Answer (5 marks each) (3 × 5 = 15)

Long Answer Questions with Internal Choice (5 marks each)
Question 1
Describe the phylum Ctenophora in detail, highlighting their distinctive characteristics. Explain how their tissue organization, locomotion mechanism, and reproductive features differentiate them from other phyla. Illustrate your answer with examples.
OR
Compare and contrast the cellular level of organization seen in Porifera with the tissue level of

organization in Coelenterates. Discuss how this difference in organization affects their physiological processes such as digestion, respiration, and reproduction.

Question 2

Explain the anatomy of a dicotyledonous stem with the help of a labeled diagram. Describe the arrangement of vascular bundles and their significance. Compare its structure with that of a monocotyledonous stem, highlighting the key anatomical differences.

OR

Describe the internal structure of a dorsiventral leaf with reference to its various tissue systems. Explain how the arrangement of tissues in the leaf is adapted for photosynthesis and gas exchange. How do these adaptations differ between dicot and monocot leaves?

Question 3

Describe the morphology and anatomy of the digestive system of a frog. Explain how its structure is adapted to its carnivorous feeding habit. Discuss the role of digestive glands in the process of digestion.

OR

Explain the reproductive system of male and female frogs. Describe the process of fertilization and development in frogs, highlighting the significance of metamorphosis in their life cycle. Discuss the ecological importance of frogs in maintaining the balance in an ecosystem.

QUESTION PAPER GENERATION SUMMARY

Class: 11

Subject: Biology

Unit: 4

Difficulty: Hard

Total Marks: 100

Duration: 3 hours

Section Details:

Section A (MCQs): 14 questions | 1 mark each | Total: 14 marks

Chapters: 4, 5 | Context: 27643 chars

Section B (Very Short Answer): 10 questions | 2 mark each | Total: 20 marks

Chapters: 5, 6 | Context: 26654 chars

Section C (Short Answer Type I): 7 questions | 3 mark each | Total: 21 marks

Chapters: 6, 7 | Context: 27717 chars

Section D (Case-based/Competency): 2 questions | 4 mark each | Total: 8 marks

Chapters: 4, 5, 6, 7 | Context: 55361 chars

Section E (Long Answer): 3 questions | 5 mark each | Total: 15 marks

Chapters: 4, 5, 6, 7 | Context: 55361 chars